

One model for every Reynolds number: the multi-regime fine-tune and the mean-dose verdict

14 August 2026. A single diffusion model was fine-tuned simultaneously against nine Reynolds numbers (1000 to 8000), each batch rewarded against its own regime's measured spectral target, to test whether one network can learn to read the flow regime from its input and adapt its output accordingly - without being told the regime. This report gives the verdict of that test and what it implies for the next step.

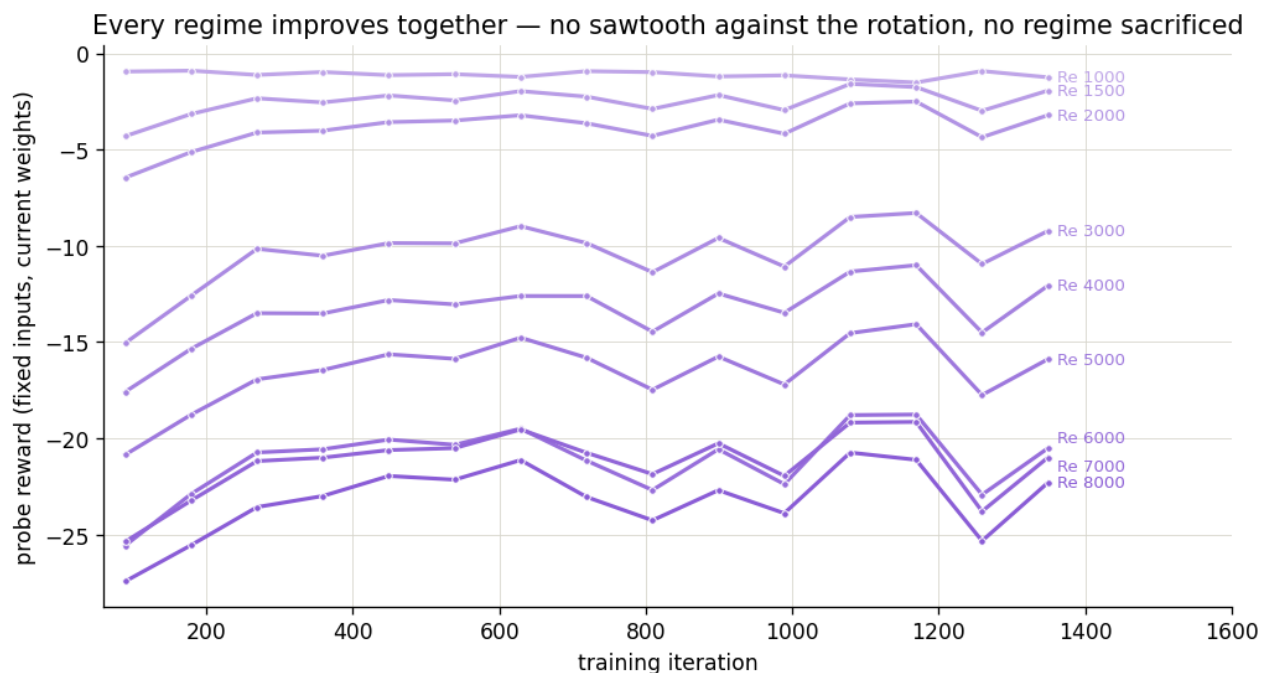
1. The question

Every single-regime fine-tune we have measured behaves the same way when moved across regimes: it restores a near-constant amount of fine-scale energy no matter what field it is given. That constant dose is exactly why transfer degrades with distance - the amount of missing energy grows with Reynolds number while the model's contribution does not. A separate study of the coarse observations showed they carry a smooth, readable regime signal (rank correlation +0.82 with the true ordering, concentrated in the spectral tail). So the question: if one model is trained on a mixture of regimes, each rewarded against its own target, does gradient pressure teach it to read that signal and scale its output dose to the input - or does it settle on one compromise dose?

2. The experiment

The base model was fine-tuned for 1350 iterations with the regime rotating every iteration (round-robin over the nine regimes, 150 visits each). Rewards used measured ground-truth spectra from each regime's training sequences - an oracle setting, deliberately not deployable, chosen to separate 'can one model learn input-dependent behaviour' from 'are the extrapolated targets accurate enough'. No architecture change was made: no regime label, no conditioning input. The network can only distinguish regimes through the fields it is given. Validation used sequences fully disjoint from training in every regime.

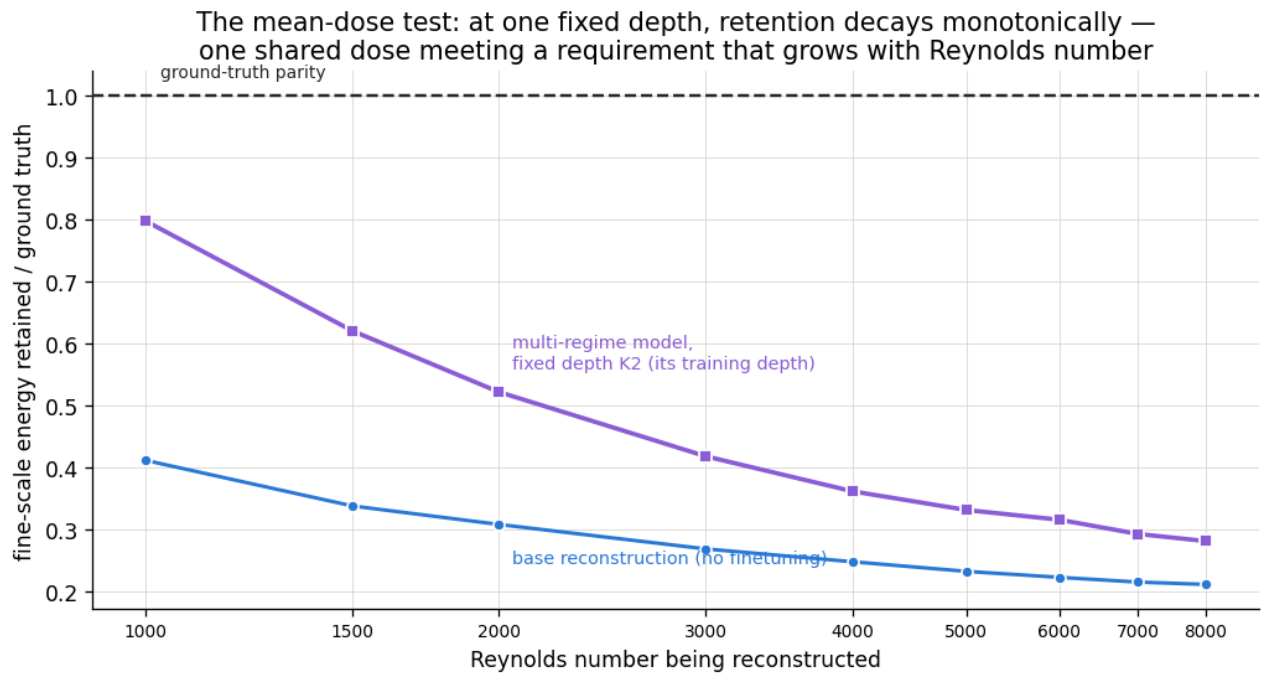
A fixed probe battery (eight held-back inputs per regime) was scored at every checkpoint. The worry this addresses: with regimes rotating, the model could oscillate - adapting to each regime in turn and forgetting the others (a sawtooth locked to the rotation).



Training was stable; every regime's probe improved together, with no sawtooth and no regime sacrificed. Note however that rising together is also what a single shared dose produces - the discriminating test follows.

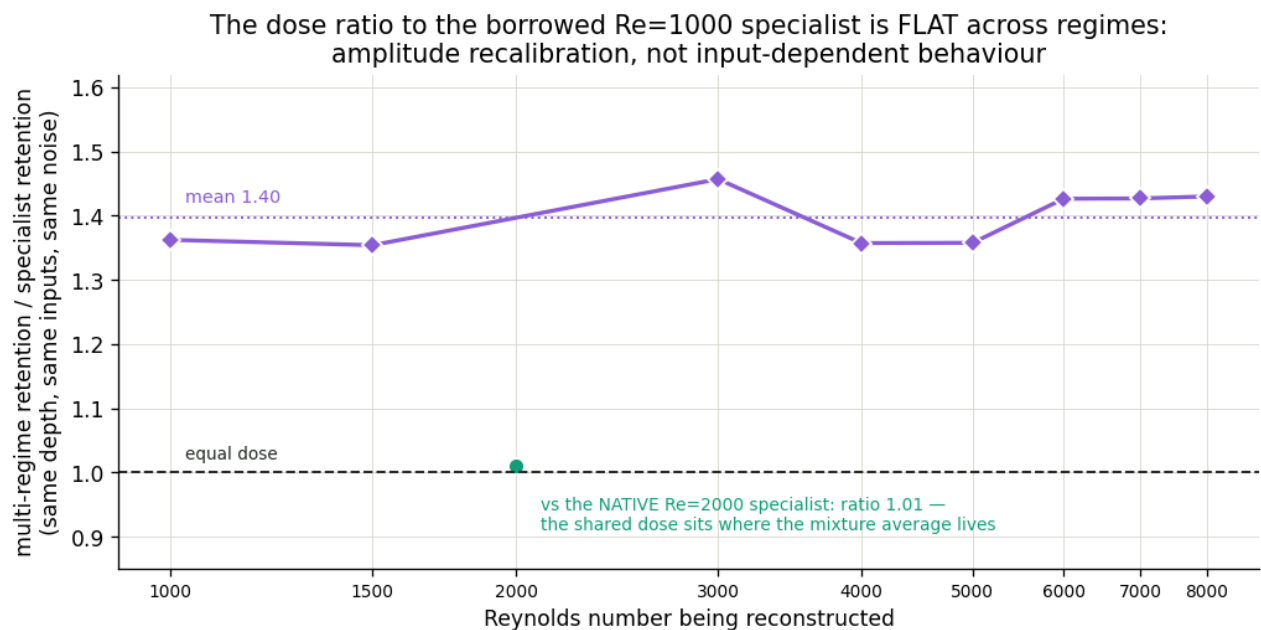
3. The mean-dose test

The decisive measurement, run before anything else on the validation sets: hold the inference depth fixed at the depth the model was trained at, and read its energy retention across all nine regimes. If the model had learned to read its input, retention would sit near 1 across the whole row. If it had learned one compromise dose, the row would decay as the ground-truth requirement grows with Reynolds number.



Retention at the fixed training depth: 0.80 at Reynolds 1000 falling monotonically to 0.28 at 8000. The decay profile of a constant dose.

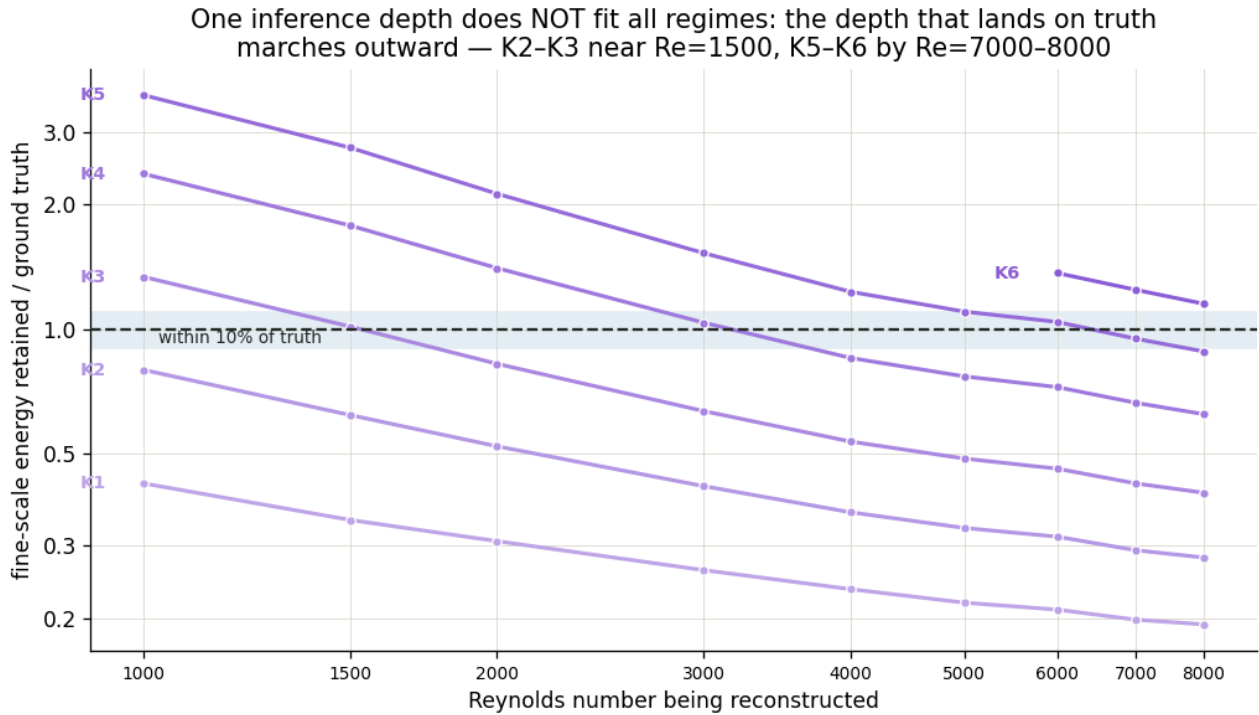
The cleanest proof that the dose is constant comes from comparing the multi-regime model with the borrowed Reynolds 1000 specialist at matched depth, on the same inputs with the same sampling noise. If the multi-regime training had produced any input dependence, the ratio between the two models' retentions would vary across regimes. It does not:



The ratio is flat at 1.40 (spread 1.35-1.46) from Reynolds 1000 to 8000. Multi-regime training recalibrated the amplitude of one shared dose - 40 percent hotter, matching the mixture's average demand - and nothing else. Against the native Reynolds 2000 specialist the ratio is 1.01: the shared dose sits almost exactly where the mixture's centre of demand lives.

4. Depth still does the adapting

Since the learned dose is one number, the only remaining dial is the inference cascade depth, which multiplies the dose by applying it over more denoising phases. Graded across the full ladder, the multi-regime model behaves exactly like the specialists: the depth that lands on the truth marches outward with Reynolds number.



Depths K2-K3 land on truth near Reynolds 1000-1500, K4 near 3000, K5 near 5000-6000, K5-K6 by 7000-8000. One configuration does not fit all regimes.

Two practical notes. First, the truth-free selection rule remains applicable to this model unchanged: its blind scores are monotone in depth exactly as for the specialists (at Reynolds 8000, for example, the blind score picks K5, which is indeed its best rung). Second, at matched depth in the far regimes the multi-regime model retains substantially more energy than the borrowed specialist (1.04 versus 0.73 at Reynolds 6000), at a modest placement cost of about 0.06 - so as a single checkpoint to carry everywhere, it is the better one; it simply is not an adaptive one. The exponential-moving-average weights carried in its checkpoints grade within noise of the raw weights.

5. Why unequal reward magnitudes did not bias the training

The raw rewards differ enormously across regimes (roughly -1 at Reynolds 1000 versus -43 at 8000 at the start of training), which raises a fair concern: does the far end steer the gradient harder? It does not, by construction. The policy-gradient update never sees raw rewards: for every input, the four sampled reconstructions are scored and their rewards standardised within that group - the group's mean is subtracted and its spread divided out - before entering the loss. Every regime therefore pushes with the same force per iteration, and only the ranking of samples within a group matters. The compromise dose is not a magnitude artefact; it is a vote-counting outcome. Seven of the nine regimes want a hotter dose than Reynolds 1000 does, the unconditioned network can only produce one dose, and the shared dose settled at the mixture's preference. Had magnitudes leaked in, the far regimes would have crushed Reynolds 1000 entirely; instead its probe held near its optimum throughout (a drift of 0.3 in 1350 iterations against gains of 5 units elsewhere).

6. Verdict and the step it dictates

Verdict: blind adaptation through mixture training does not produce input-dependent behaviour in this architecture. The regime signal is present in the input, but nothing in the network's structure gives it a cheap

pathway from that signal to a global output scale, and 150 visits per regime of policy-gradient pressure did not build one.

The step this dictates is explicit conditioning: give the network the regime as an input and let the reward shape what it does with it. Concretely - a FiLM-style pathway: a small embedding of the (always known) target Reynolds number added to the existing time-embedding vector, which already modulates every residual block through per-channel scales and shifts. The new pathway is initialised at exactly zero, so the conditioned model starts as a bit-identical copy of the base model and the fine-tune only has to learn the modulation, not relearn reconstruction. The same nine-regime rotation, rewards and probe battery then re-run unchanged; the mean-dose row of section 3 is the acceptance test. If the conditioned row is flat where the unconditioned row decayed, the conditioning works.

Provenance: training script src/ddpo_ft/multiregime_ft.py (checkpoint directory monitoring/ddpo_multiregime_gt_ckpts, synced to cloud storage); grading harness src/ddpo_ft/multiregime_grade.py (results base_results/multiregime_grade.npz); figures monitoring/plot_multiregime_verdict.py. Validation pools: generated-regime sequences 8-19 and Reynolds-1000 sequences 4-19, disjoint from all training data.